

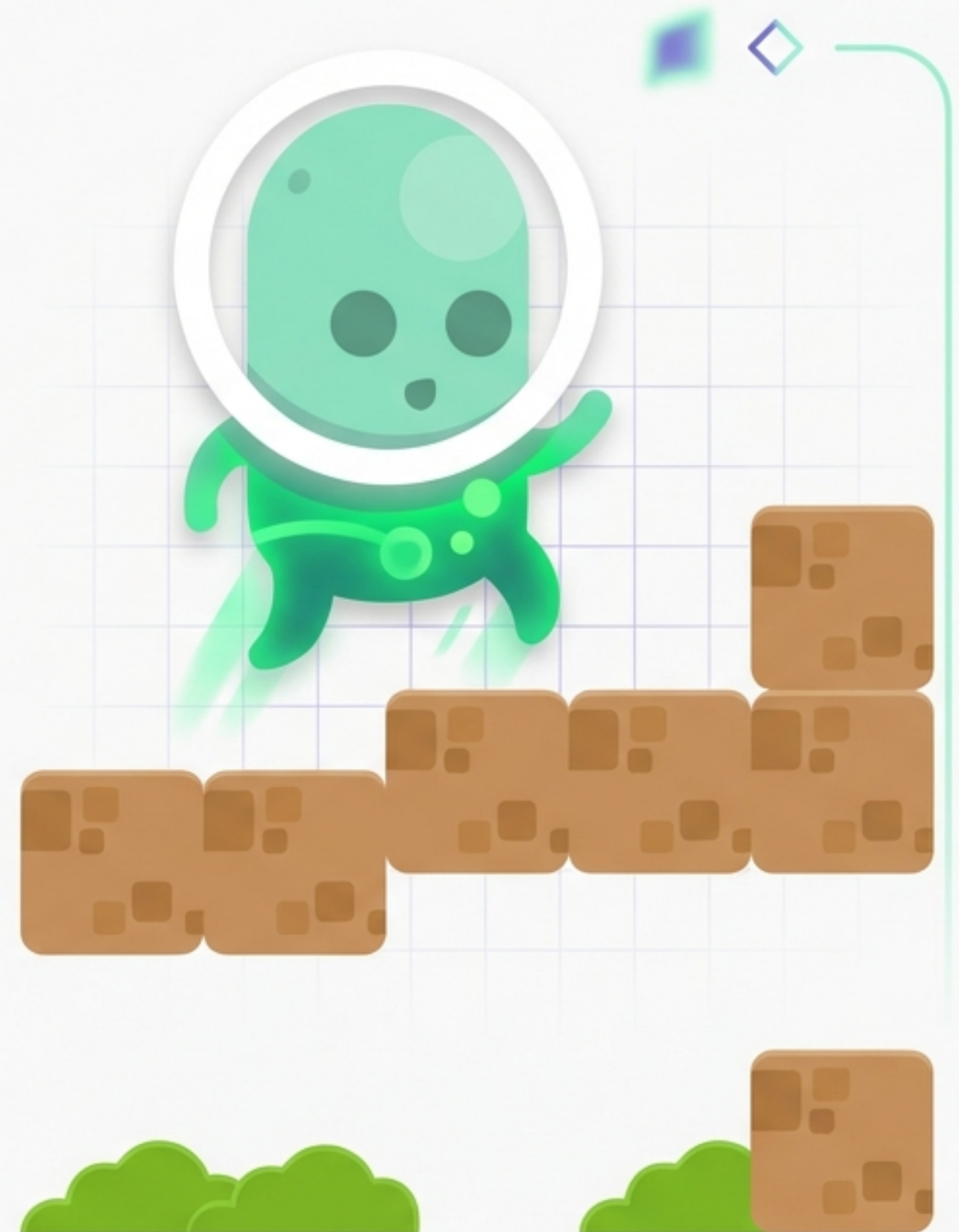
Level Up: Platformer Game Mechanics

ICT 360 | Grade 6 | Lesson 11

Loading...

✓ 100% COMPLETE

In the last lesson, we made a square move.
Today, we turn that square into a hero.



Today's Mission Objectives



Life in Motion

Stop the "statue sliding" by adding Animations.



```
player.AddAnimation("run");
```



Better Flow

Stop hitting your head on blocks by using Jumpthru Platforms.

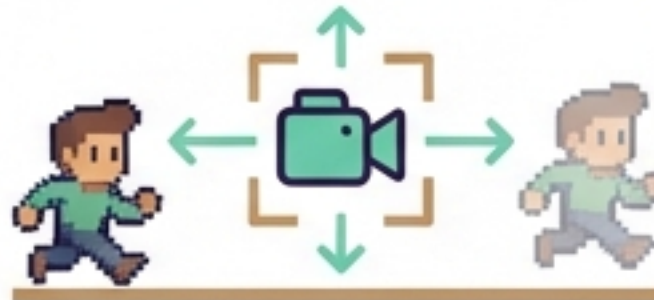


```
platform.SetJumpthru(true);
```



Open World

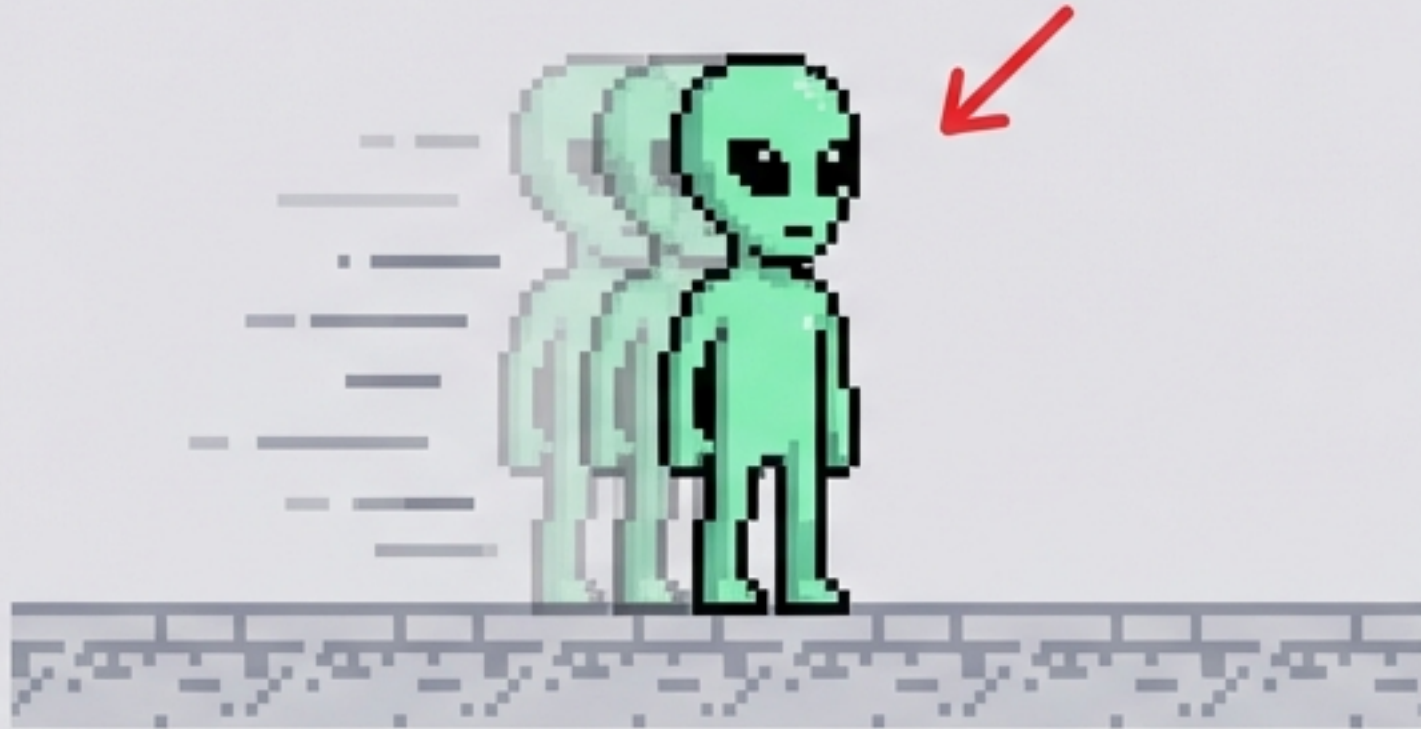
Stop getting lost off-screen by adding Camera Follow.



```
camera.EnableFollow(player);
```


The Problem: Is Your Hero Sliding Like a Statue?

Just Moving (Boring)

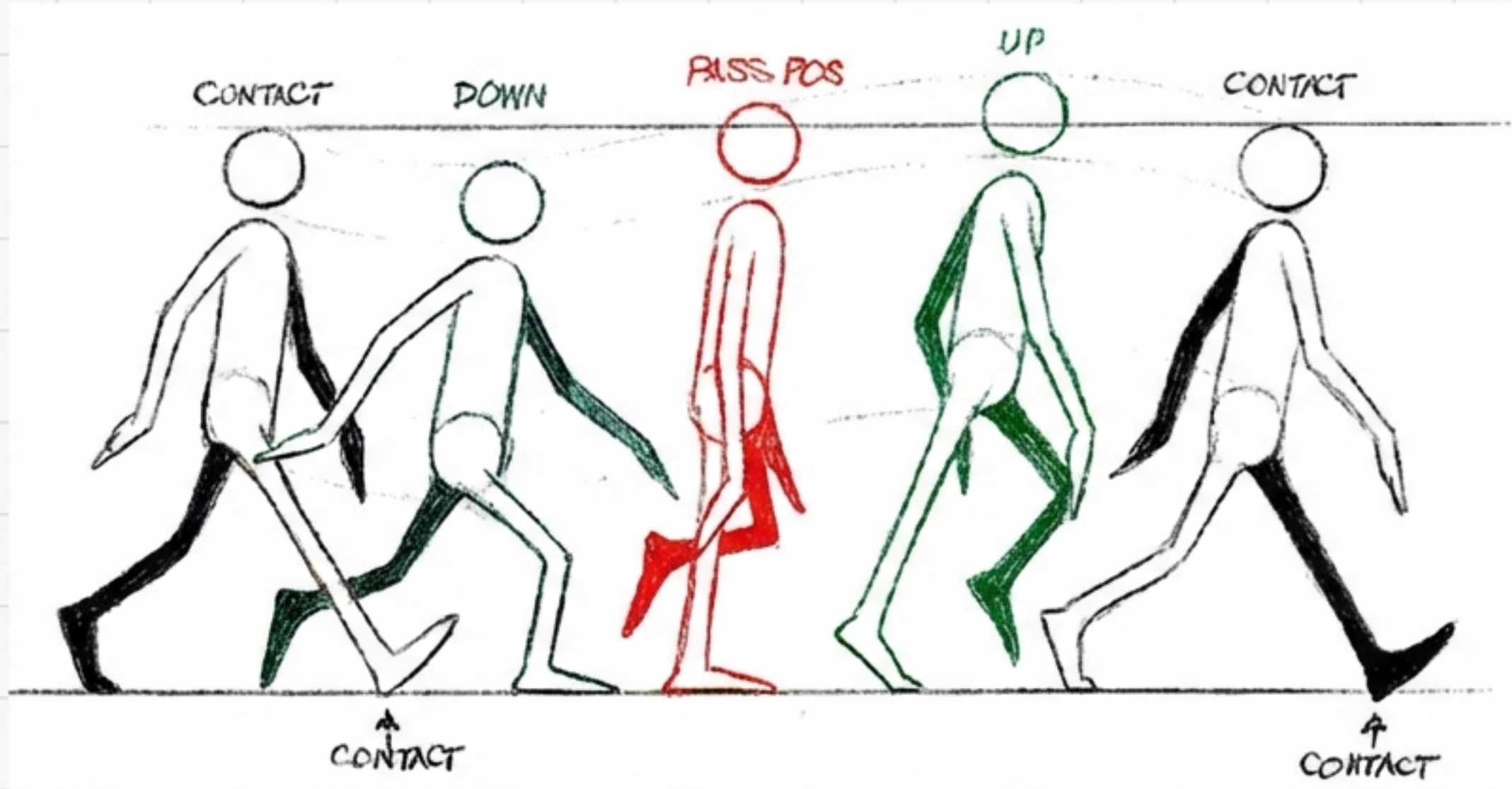


Animating (Alive)



We programmed movement, but our character looks frozen. Real actions—walking, talking, running—are never static. Just like our lips move when we talk, a sprite needs to cycle through images to look real.

The Solution: The Walk Cycle



Animation is a trick of the eye. It is a series of illustrations (frames) played in a specific order to create the illusion of movement.

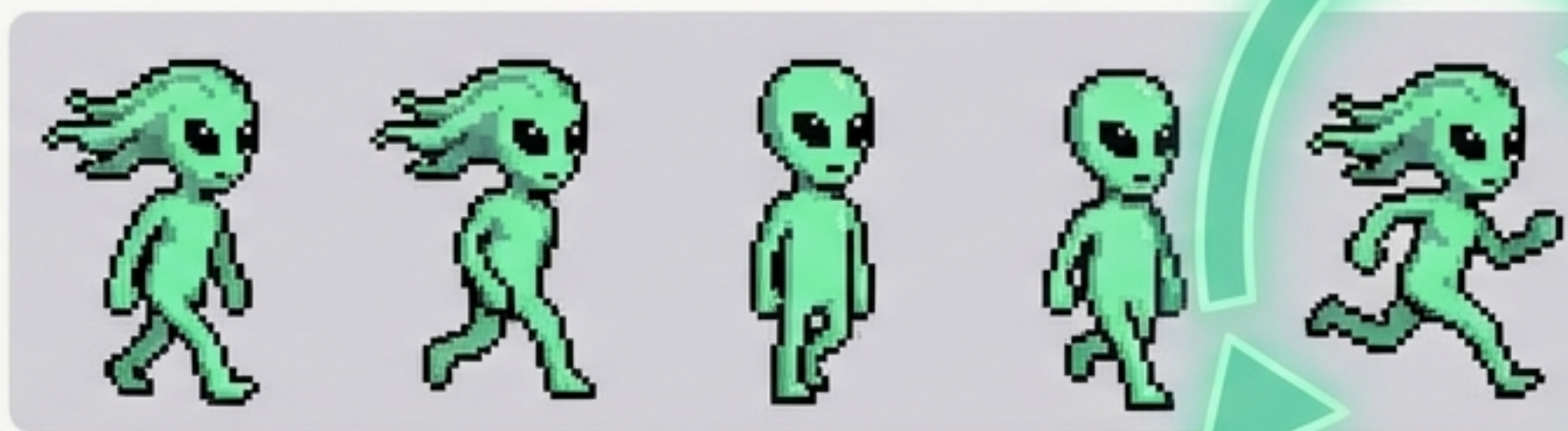
Key Concept:

We don't animate every single step. We create one 'Cycle' and repeat it.

Mechanics: The Infinite Loop



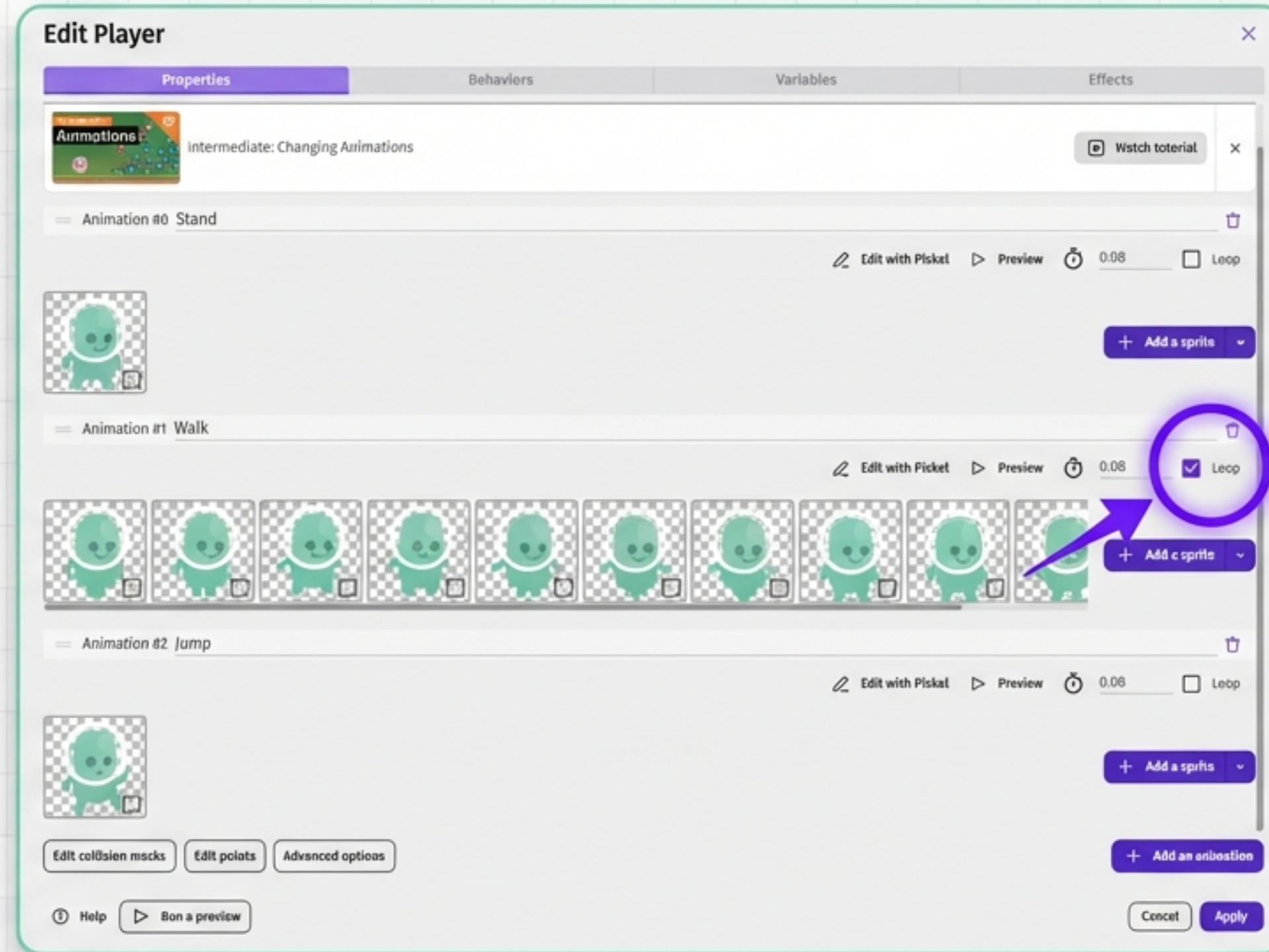
The Clock Rule: The hands rotate 1 through 12, then immediately start over at 1. Your animation frames must do the same.



Loop

Setting Up the Loop

1. Open your Player object.
2. Locate the 'Walk' animation.
3. Check the Loop box.



Pro Tip: Notice the Timer icon? That controls how fast the frames flip!

Naming Your Moves



Animation #0 -> "Stand"



Animation #1 -> "Walk"



Animation #2 -> "Jump"

The computer needs to know exactly which animation to play and when.
Always name your animations clearly in the properties panel.

Why?

If you don't name them, you have to remember them by number (0, 1, 2), which gets confusing!

Coding the Behavior



Player is jumping Add condition	Change the animation of Player : set to "Jump" Add action
Player is falling Add condition	Change the animation of Player : set to "Jump" Add action
Player is on floor Add condition	Add action
Player is moving Add condition	Change the animation of Player : set to "Walk" Add action
Player is moving Add condition	Change the animation of Player : set to "Stand" Add action

Condition

Condition: Player is moving ->
Action: Set animation to "Walk"

Condition

Condition: Player is jumping ->
Action: Set animation to "Jump"

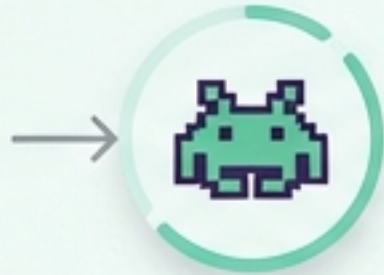
JetBrains Mono

Condition: Player is still ->
Action: Set animation to "Stand"

Face the Right Way

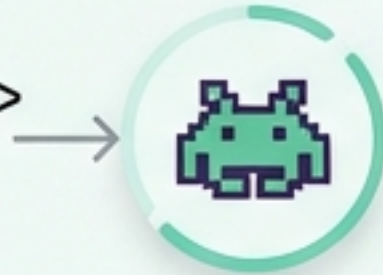
Condition → Action

Left key is pressed ->
Flip horizontally: YES



Condition → Action

Right key is pressed ->
Flip horizontally: NO



Explanation

Animations don't automatically turn around. We must flip the sprite horizontally to make it face the direction of travel.

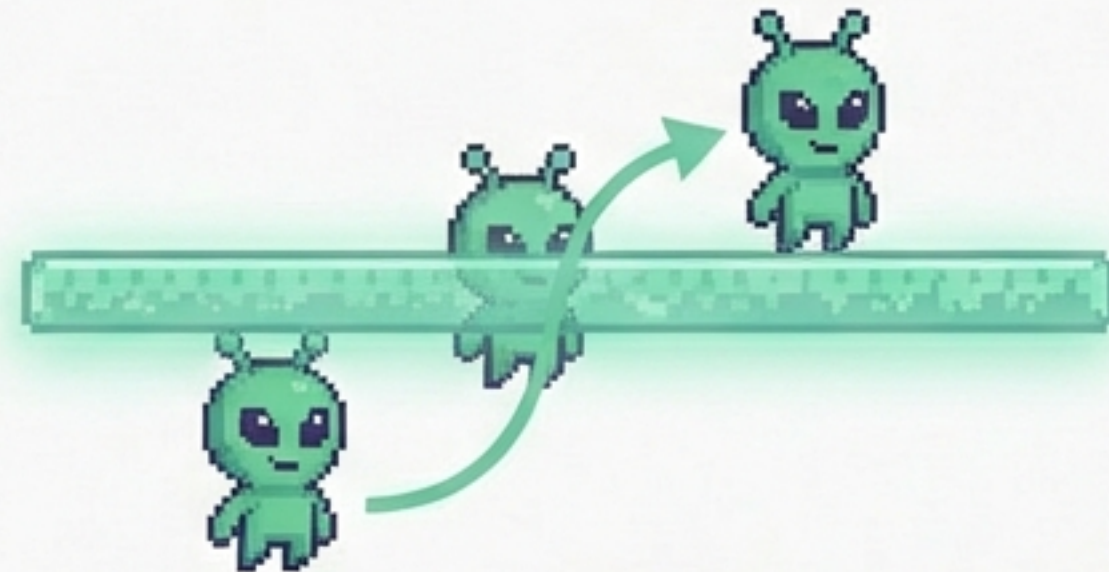
Problem #2: Blocking the Flow

Solid Platform



Character cannot pass through from below.

Jumphru Platform



Character passes through from below and lands on top.

Jumphru

A behavior that allows the player to jump onto the platform from beneath it, but land solidly on top of it.



Creating a Jumpthru Platform

Setup Steps



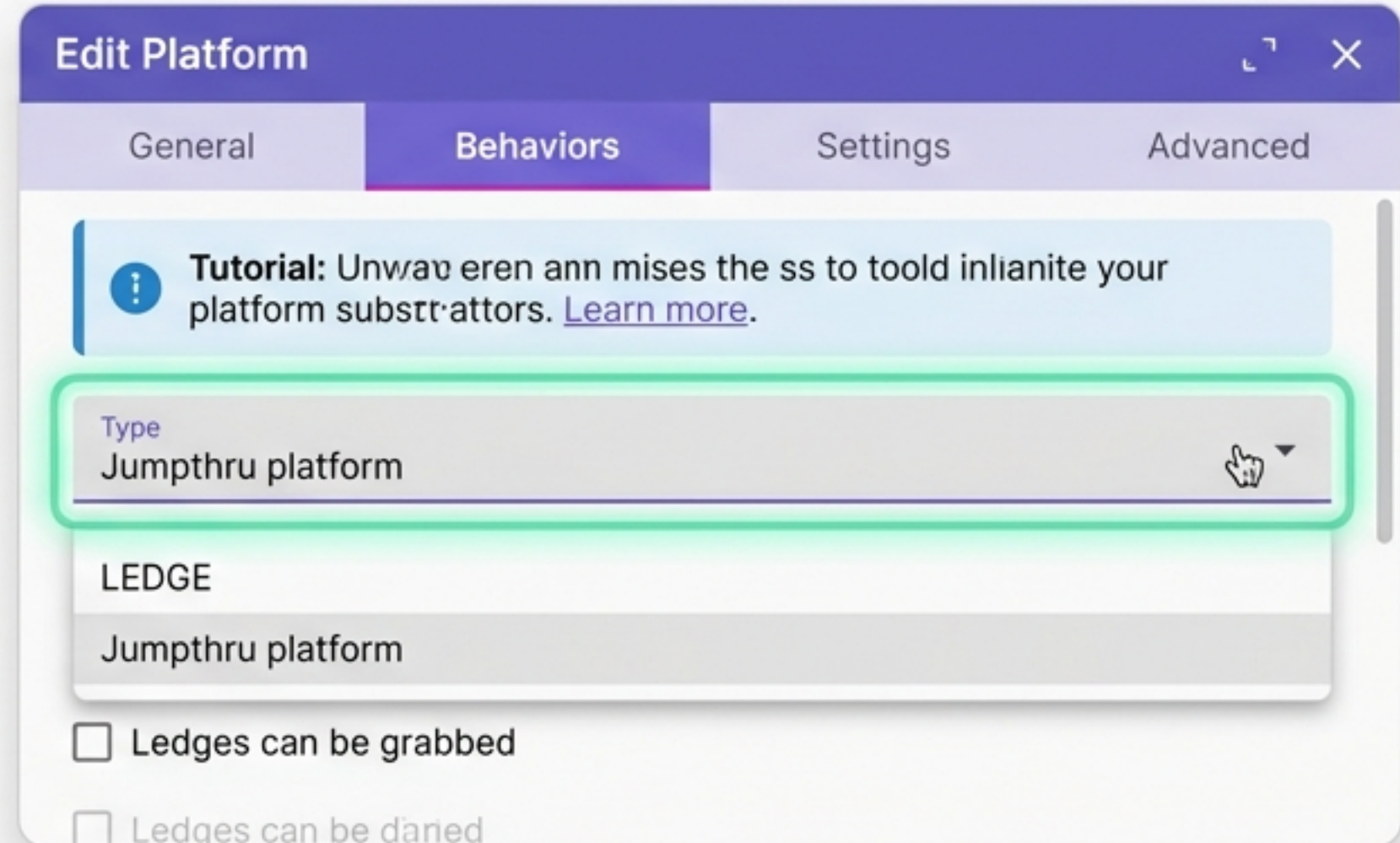
1. Create a new Sprite object.



2. Add the Platform behavior.



3. Change the Type to "Jumpthru platform".



Result

Your player can now ascend the level smoothly without getting blocked.

Problem #3: Lost in Space



Where did he go?

Game levels are usually much wider than the screen window. If the player walks right, they leave the visible area.

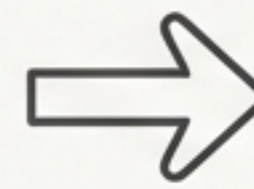
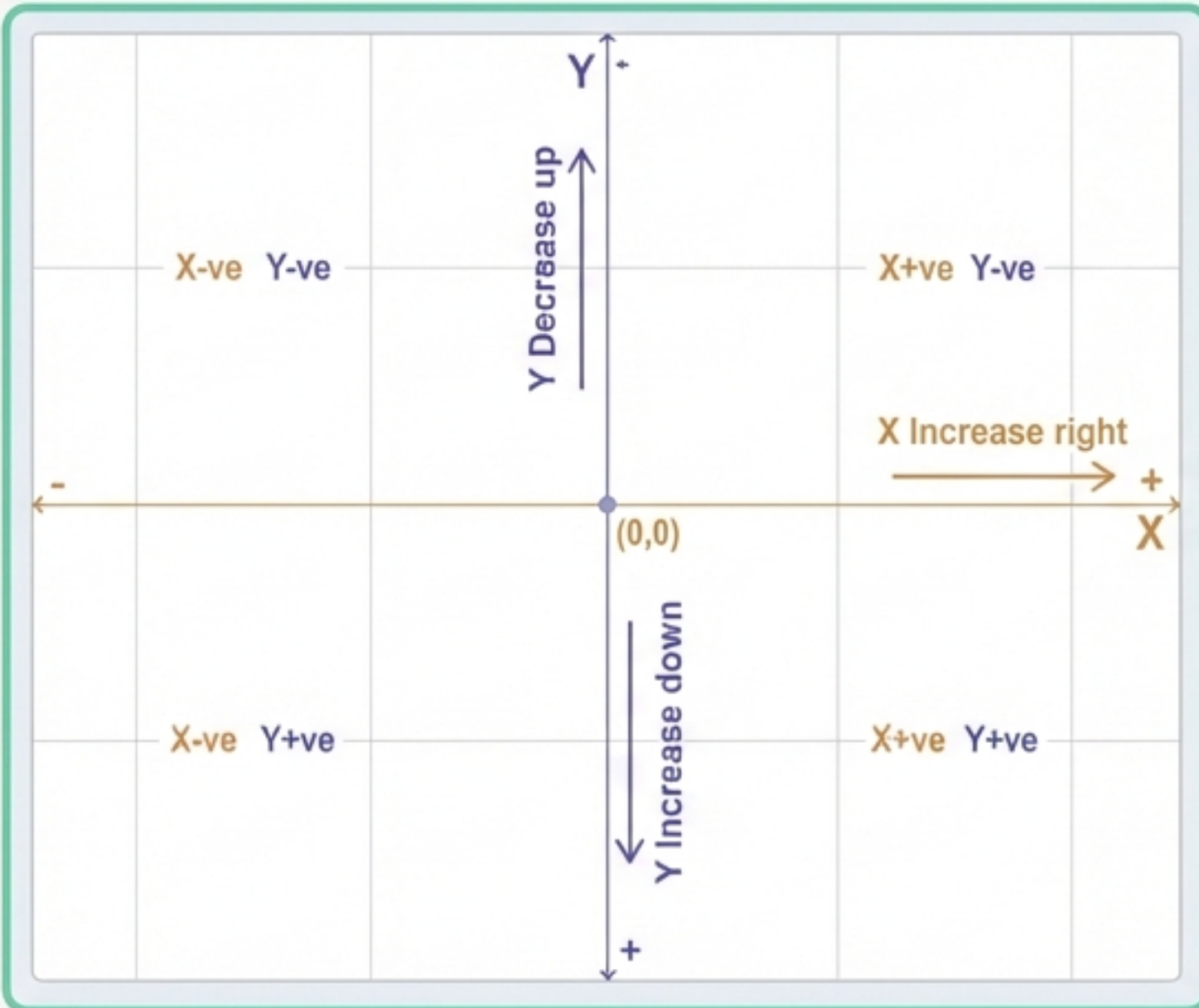
We need a camera operator!



Camera Follows
Player



Navigation: The Coordinate System



X-Axis:
Left (-) to Right (+)



Y-Axis:
Up (-) to Down (+)



Watch Out! !

In GDevelop, the Y number gets BIGGER as you go DOWN!

Locking the Camera

📹 Change the X position of camera (layer: Base layer): `set to Player.X()`

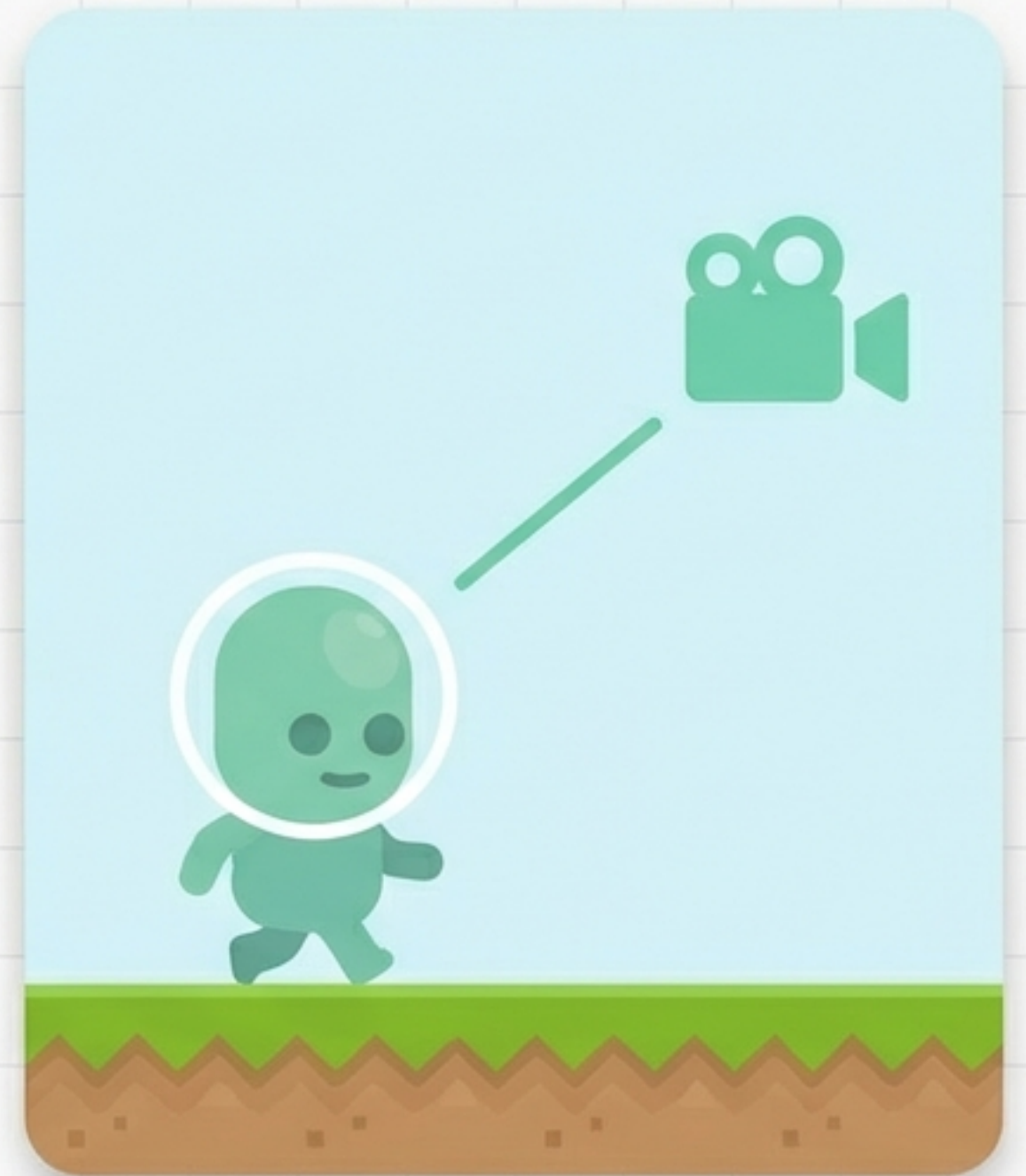
The Logic

Wherever Player X goes, Camera X follows.



Tech Flash

This action executes every single frame (60 times a second) to keep the view smooth.



Mission Debrief: Tech Flash

Inventory

Execute Every Frame



An action that runs continuously (60 times/sec) to ensure smooth updates.

Jumpthru Platform



A specific behavior allowing players to pass through a platform from below.

Animation Name



The ID used in code (e.g., "Walk" in **JetBrains Mono**) to switch visual states via Events.

Final Challenge: Now it's your turn. Add obstacles, gravity, and design your own level!